All-real Jacobians and exact response fibers of finite tree compartment models

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Abstract

For a finite labelled bidirected tree with collocated input and state observation at o and one external leak at any vertex ℓ, we factor the actual characteristic/cofactor coefficient Jacobian at every real parameter vector. The factors are one rate on each edge directed toward ℓ, each edge product to its observed-subtree-size-minus-one power, and the squared resultants of the actual sibling principal polynomials. Signed triangular witnesses prove every coordinate multiplicity is exact, including multiplicity zero for the leak, and identify the entire real Jacobian zero set. The root-leak specializations imply the catenary and tree-multiplicity conjectures of Gross, Meshkat and Shiu. For every full positive Stieltjes response, we also classify all real realizations: positivity and Jacobian regularity are consequences of the response, and the exact count is $(N-1)!/\prod_{v\neq o}|T_v|$ at each prescribed leak site. An unknown site multiplies this actual matrix count by N. We supply ordinary proofs and a map to a fixed, independently kernel-checked Lean package. Verification certifies the specified statements; historical priority and external peer review are separate questions.

1 Context and precise contribution

Identifiability of linear compartmental systems has a substantial classical literature. Tree-type realization counts were already investigated by Cobelli, Lepschy and Romanin Jacur [1]; we make no claim to have invented tree enumeration or Stieltjes reconstruction. Gross, Meshkat and Shiu [2] posed a path singular-locus monomial conjecture (Conjecture 5.5) and a rooted-tree coordinate-multiplicity conjecture (Conjecture 5.9), the latter attributed there to Hoch, Sweeney and Tung. Our factorization and exact-power statements imply both in their original collocated-leak setting. Ahmed et al. [3], Section 6, still discuss the earlier path conjecture as a prospective application of their coefficient formulas. These correspondences do not establish that the present proof is historically first: targeted searches cannot rule out all later or unpublished work.

The paper's main algebraic result includes arbitrary single-leak location, every real rate, the full sibling-resultant residual, and exact coordinate powers. A second result gives an exhaustive real fiber for each full distinct-positive-pole target, with automatic admissibility of every real competitor. It distinguishes this non-generic target statement from a generic complex identifiability degree. The matrix, coefficients, derivatives and ODE observations are the actual compartment objects throughout.

2 Actual model and all-real factorization

Let $T = (V, E)$ be a finite labelled simple tree, $|V| = N \geq 1$. Fix the collocated input and state observer $o \in V$, and one leak vertex $\ell \in V$. For each directed edge $u \to v$ let $k_{vu} \in \mathbb{R}$ be its transfer rate, and let $\lambda \in \mathbb{R}$ be the external leak at ℓ. Define

$$A_{vu} = k_{vu} \quad (u \sim v), \qquad A_{uu} = -\sum_{v \sim u} k_{vu} - \lambda \, \mathbf{1}_{u=\ell}, \tag{1}$$

with all other off-diagonal entries zero. Thus $\mathbf{1}^T A = -\lambda e_\ell^T$. The dynamics are $\dot{x} = Ax + e_o u$, with state output $y = e_o^T x$.

Put

$$P(s) = \det(sI - A), \qquad F_o(s) = \det((sI - A)[V \setminus \{o\}]), \qquad G_o(s) = \frac{F_o(s)}{P(s)}. \tag{2}$$

The principal submatrix retains its original diagonal entries, including rates exiting the retained set. The empty determinant is one. The two polynomials are monic of degrees N and $N - 1$. No cancellation is performed before defining the coefficient map.

Root the tree at o. For $v \neq o$, let $p(v)$ be its parent, T_v its descendant vertex set, $N_v = |T_v|$, $a_v = k_{p(v),v}$ and $f_v = k_{v,p(v)}$. Fix a parent-before-child enumeration and use parameter order $(\lambda, (a_v)_{v \neq o}, (f_v)_{v \neq o})$. The coefficient output is

$$C_{o,\ell} = (P_0, \ldots, P_{N-1}, (F_o)_0, \ldots, (F_o)_{N-2}),$$

in ascending powers; monic leading coefficients are omitted. This is a square polynomial map in $2N - 1$ real variables. Write $J_{o,\ell} = \det DC_{o,\ell}$. The Lean graph entry uses the same ascending array followed by a fixed coordinate permutation; transporting between conventions changes only a rate-independent sign. It never introduces a parameter-dependent factor.

For every child c of any observed-root vertex v, define the actual principal polynomial

$$P_c(s) = \det(sI - A[T_c]).$$

Define the observed sibling residual and leak-oriented product by

$$S_{o,\ell} = \prod_{v \in V} \prod_{\{c,d\} \subseteq \mathrm{child}(v)} \mathrm{Res}(P_c, P_d)^2, \qquad L_\ell = \prod_{e \in E} k_{e \to \ell}. \tag{3}$$

Here $k_{e \to \ell}$ is the rate on e pointing toward ℓ along its unique tree path. Empty products are one. Each P_c, including its diagonal, depends on the original model with leak at ℓ.

Theorem 1 (All-real Jacobian and exact powers)**.** *There is a sign $\varepsilon_o \in \{1, -1\}$, depending only on the chosen observation-root coordinate convention, such that simultaneously for every leak site ℓ,*

$$J_{o,\ell} = \varepsilon_o L_\ell \prod_{v \neq o} (a_v f_v)^{N_v - 1} S_{o,\ell}. \tag{4}$$

This is an identity of multivariate real polynomials and of the actual Fréchet derivative determinant at every real vector. The exact power of either directed coordinate on the edge at v is

$$N_v - 1 + \mathbf{1}_{\textit{that direction points toward } \ell}. \tag{5}$$

The leak coordinate has exact power zero. Hence $J_{o,\ell} = 0$ precisely when a rate with positive displayed exponent vanishes, or $S_{o,\ell} = 0$. Zero leak, zero edge rates, repeated spectra and signed parameters are included.

Exact power means that the stated power of the coordinate variable divides the formal Jacobian polynomial and the next power does not. The residual is not asserted irreducible. The zero set is a Jacobian-rank statement; it need not characterize every notion of local uniqueness at singular points.

Corollary 2 (Rooted trees and endpoint paths)**.** *For $\ell = o$, the inward and outward coordinate powers are respectively N_v and $N_v - 1$. On the path $1, \dots, N$ with $o = \ell = 1$,*

$$J = \varepsilon_o \prod_{i=1}^{N-1} k_{i,i+1}^{N-i} k_{i+1,i}^{N-i-1}.$$

These give the path monomial and rooted-tree multiplicities of [2, Conjectures 5.5 and 5.9], in the stated original coordinate convention up to sign.

For the path there are no sibling pairs, so $S = 1$. The rooted-tree powers satisfy the subtree recursion $N_v = 1 + \sum_{c \in \mathrm{child}(v)} N_c$, with the corresponding inward/outward leaf values $1, 0$. This is the required multiplicity implication, rather than mere divisibility by a candidate monomial.

3 Proof of the polynomial theorem

3.1 Coefficient assembly and the root-leak case

The following elementary determinant calculation is the algebraic source of the squared sibling resultants.

Lemma 3 (Polynomial gluing)**.** *Let p_i be monic of degree $n_i \geq 1$, let q_i have degree less than n_i, and let d be a scalar. With $K = \sum_i n_i$, put*

$$F = \prod_i p_i, \qquad P = (s + d)F - \sum_i q_i \prod_{j \neq i} p_j.$$

The map from the non-leading coefficients of the p_i, all coefficients of the q_i, and d, to the non-leading coefficients of F, P has determinant

$$\sigma \prod_{i<j} \mathrm{Res}(p_i, p_j)^2,$$

where $\sigma = \pm 1$ depends only on the fixed coefficient ordering. This identity holds without coprimality or simple-root hypotheses.

Proof. Write $\mathcal{L}_p(h_i) = \sum_i h_i \prod_{j \neq i} p_j$, with $\deg h_i < n_i$. This is a square linear map to polynomials of degree below K. The derivative of $(p_i) \mapsto F$ is exactly $\mathcal{L}_p$. In coefficient bases its determinant is a fixed sign times $\prod_{i<j} \mathrm{Res}(p_i, p_j)$.

Here is a direct verification of the latter assertion. Initially take all roots to be simple and mutually disjoint in $\mathbb{C}$. Evaluate at the roots of each p_i, in fixed block order. Only h_i contributes at its own block, with diagonal multipliers $\prod_{j \neq i} p_j(\alpha)$. The quotient of the full Vandermonde determinant by the within-block Vandermondes is the cross-root difference product. Cancelling one copy of that product in the evaluation identity leaves one resultant for each pair, with a fixed ordering sign. Both sides are polynomials in the coefficients, so equality on this nonempty open set proves the identity for every coefficient vector.

For fixed (p_i), variations of (q_i, d) give $\delta P = \delta d\, F - \mathcal{L}_p(\delta q_i)$. Since F is monic, its s^K coefficient supplies the δd pivot; the remaining determinant is the same $\det \mathcal{L}_p$ up to a fixed sign. The entire derivative is block triangular because F does not depend on q_i, d. Its determinant therefore contains two copies of the resultant product. This also proves the empty-family case $F = 1, P = s + d$. □

Suppose first that the leak is at o. Its child branches have sizes n_i, actual characteristic polynomials P_i and root cofactors F_i. The principal branch is a genuine child model whose leak is the inward connector a_i. Original diagonals are essential here. Schur expansion, as a polynomial identity, gives

$$F_o = \prod_i P_i, \qquad P = (s+d)\prod_i P_i - \sum_i a_i f_i F_i \prod_{j\neq i} P_j, \quad d = \lambda + \sum_i f_i. \tag{6}$$

It can first be checked where the branch blocks are invertible and then extended polynomially; it asserts an identity of actual determinants.

Keep the child rate vectors, including their leak a_i, and change the remaining variables from $(\lambda, (f_i))$ to $(d, (a_i f_i))$. The determinant is $\prod_i a_i$, by its block triangular derivative, at all rate values. Scaling the monic F_i by $b_i = a_i f_i$ changes $(b_i, \operatorname{coef} F_i)$ into $\operatorname{coef}(b_i F_i)$ with determinant $b_i^{n_i-1}$ up to a fixed sign, even at $b_i = 0$. Applying Lemma 3 and each actual child Jacobian gives

$$J_T = \sigma_T \prod_i a_i (a_i f_i)^{n_i-1} J_{T_i} \prod_{i<j} \operatorname{Res}(P_i, P_j)^2. \tag{7}$$

All coordinate changes here are explicitly described; no inverse rate is used. Induction, starting with $P = s + \lambda, F = 1, J = 1$, flattens (7) into (4) for $\ell = o$. The sign depends only on the rooted enumeration and the fixed permutations.

3.2 Leak relocation and removal of denominators

On the positive orthant, move a leak λ at v across an adjacent edge to u. Put $a = k_{uv}$ and $f = k_{vu}$. Replace only these three coordinates by

$$a' = a + \lambda, \qquad f' = \frac{af}{a+\lambda}, \qquad \lambda' = \frac{f\lambda}{a+\lambda}, \tag{8}$$

with the new leak at u. The diagonal entries stay the same, since $f' + \lambda' = f$ and $a' = a + \lambda$. Cutting that edge gives two components. A diagonal scaling constant on each component, with ratio $D_v/D_u = a/(a+\lambda)$, gives $A' = DAD^{-1}$. It preserves every principal characteristic polynomial and every edge product. In particular it preserves both P, F_o and the entire core $W = \prod_{v\neq o}(a_v f_v)^{N_v-1} S_{o,\ell}$.

A direct derivative of (8), in order (λ, a, f) and (λ', a', f'), has determinant $f/(a+\lambda) = \lambda'/\lambda$. Compose the slides on the unique path from ℓ to o. If R is the resulting positive rate transformation and λ_o its new leak,

$$C_{o,\ell} = C_{o,o} \circ R, \qquad \det DR = \lambda_o/\lambda.$$

The root-leak factorization and the chain rule imply

$$J_{o,\ell} = \varepsilon_o \frac{\lambda_o}{\lambda} F_o^{\text{new}}(0) W. \tag{9}$$

For any single-leak tree model, conservation and expansion in the leak give

$$P(0) = \lambda F_\ell(0), \qquad F_\ell(0) = L_\ell. \tag{10}$$

For completeness, the leak-free matrix has zero determinant because its columns sum to zero, and the leak enters a single diagonal linearly. For the second identity root at ℓ: eliminating a nonroot leaf in the positive model contributes its inward rate to the principal determinant, and its Schur correction cancels its outward contribution to the parent diagonal. Iterating gives exactly L_ℓ. Since these are polynomial identities, they hold at every real specialization as well.

The characteristic polynomial is invariant under R, so $\lambda_o F_o^{\mathrm{new}}(0) = \lambda F_\ell(0)$. Substitute into (9). The result is $J_{o,\ell} = \varepsilon_o F_\ell(0) W$, with no transformed rate or denominator. Both sides are polynomial functions of the original coordinates. Equality on the product of positive half-lines proves polynomial equality, by applying the one-variable infinite-zero principle successively to each coordinate. Thus (4) holds on all of $\mathbb{R}^{2N-1}$, with the same sign for every ℓ. We have not divided at a zero-boundary point or asserted that the positive orthant is dense in the entire parameter space.

3.3 Exact coordinate powers

Fix a rate coordinate, or the leak coordinate. Root the tree instead at ℓ, set all rates pointing away from ℓ to zero, and assign numbers b_v to the inward rates at nonroot vertices and b_ℓ to the leak. Choose all b_v distinct. Then A is triangular in leak-root order, with diagonal $-b_v$. Every principal characteristic polynomial is the product of the diagonal factors for its original vertices, even when that principal set is defined in observation-root order. Sibling sets are disjoint; therefore all their resultants are nonzero, so $S_{o,\ell} \neq 0$.

This witness can put the selected coordinate on its zero hyperplane. An outward-from-leak rate is already zero. For an inward coordinate at v, set $b_v = 0$ and choose every other b_w distinct and nonzero. For the leak coordinate set $b_\ell = 0$ instead. Signed choices are allowed because the claim is algebraic. Thus no coordinate variable divides the residual polynomial. Each coordinate variable is prime in the multivariate polynomial ring, so its exact power in (4) is the displayed monomial exponent. The leak has exponent zero. Since the sign is nonzero, the stated real zero set follows directly from the product. This finishes the proof of Theorem 1.

4 Full positive targets and exhaustive real fibers

Fix

$$G(s) = \sum_{i=1}^{N} \frac{w_i}{s + r_i}, \qquad r_i > 0 \text{ distinct}, \quad w_i > 0, \quad \sum_i w_i = 1. \tag{11}$$

Equality of responses means equality on the common real nonpole domain, equivalently rational-function equality. A real competitor has the fixed labelled tree T and one prescribed leak site, with no sign, stability, minimality or Jacobian condition imposed beforehand.

Theorem 4 (Complete real full-response classification)**.** *For each prescribed ℓ, every real realization of (11) has strictly positive directed rates and leak and a nonzero actual coefficient Jacobian. Its entire fiber is in bijection with recursive labelled pole allocations and has cardinality*

$$D(T, o) = \frac{(N-1)!}{\prod_{v \neq o} N_v}. \tag{12}$$

If the leak site is unknown, the fiber contains exactly $ND(T, o)$ distinct physical matrices. Labels are retained; tree automorphisms are not quotiented. The singleton case has one realization. These conclusions hold for every target (11), not only a generic subset of that class.

4.1 Reciprocal data and positive reconstruction

Order $r_1 < \cdots < r_N$. On each interval between consecutive negative poles, $G'(s) = -\sum_i w_i/(s + r_i)^2 < 0$; its one-sided limits have opposite signs. Thus its numerator has exactly $N - 1$ simple negative zeros $-\beta_j$. Partial fractions give the unique identity

$$\frac{1}{G(s)} = s + d - \sum_{j=1}^{N-1} \frac{\gamma_j}{s + \beta_j}, \qquad \beta_j > 0, \quad \gamma_j = -\frac{1}{G'(-\beta_j)} > 0. \tag{13}$$

The monic degrees determine the linear polynomial part. In particular

$$\lambda = d - \sum_j \gamma_j/\beta_j = 1/G(0) > 0.$$

For root-child sizes n_c, allocate the $N-1$ reciprocal poles into labelled disjoint sets I_c of those sizes. Put

$$b_c = \sum_{j\in I_c} \gamma_j, \qquad G_c(s) = \sum_{j\in I_c} \frac{\gamma_j/b_c}{s+\beta_j}.$$

Recursively realize G_c on its labelled child tree. Its root leak is $a_c = 1/G_c(0) > 0$. Choose $f_c = b_c/a_c = \sum_{j\in I_c} \gamma_j/\beta_j$. The parent leak is $1/G(0)$, and its full root diagonal is $\lambda + \sum_c f_c = d$. Formula (6) recovers exactly the target characteristic polynomial and cofactor. This constructs a positive realization for every recursive allocation.

4.2 Reverse extraction and the entire root-leak fiber

Write the target monic denominator and numerator as P, F. They are coprime: at the pole $-r_i$, the numerator is $w_i \prod_{j\neq i}(r_j - r_i) \neq 0$. For any N-vertex real competitor with equal response, cross multiplication gives $F_x P = F P_x$. Coprimality and the equal monic denominator degrees force $P_x = P$ and $F_x = F$. Coefficient equality is therefore recovered from the original response, not additional observed information.

When the leak is at o, its actual child denominators are monic and multiply to $F = \prod_j (s+\beta_j)$. They must partition these distinct linear factors in exactly one labelled root allocation. In (6), compare their branch numerators with those of the positive reference for that allocation. Reducing the numerator difference modulo each P_c isolates its term; the product of the other, coprime denominators is invertible modulo P_c. The term has degree below n_c, so it must be zero. Thus $a_c f_c F_c$ equals the reference branch numerator. Its leading coefficient gives $a_c f_c = b_c > 0$ and its monic cofactor is the reference cofactor. Each actual child consequently has the full target G_c. Induction forces its real rates and its leak a_c positive; then $f_c = b_c/a_c > 0$, and the parent leak is $1/G(0) > 0$. This is a proof about every real competitor.

Forward assembly and reverse extraction are inverse on actual rate vectors. There are $(N-1)!/\prod_c n_c!$ labelled root allocations; permutations within a block do not make new allocations. Hence

$$D(T,o) = \frac{(N-1)!}{\prod_c n_c!} \prod_c D(T_c, c).$$

Induction gives (12), with finiteness established by the explicit bijection. Different sibling branches receive disjoint poles at each depth, so every sibling resultant is nonzero. Theorem 1 now gives actual Jacobian regularity for all these realizations.

4.3 Automatic positivity at an arbitrary leak site

One must not move the leak of a signed competitor using positive-only formulas before proving admissibility. We instead use the following diagonal-gauge construction for general real tree-supported matrices.

Let M have this tree support and the full pair P, F at o; no conservation or sign condition is imposed. The same child factor partition and partial-fraction uniqueness recover its root diagonal, child cofactors, and every boundary edge product $M_{oc}M_{co} = b_c > 0$. Inductively each actual child matrix is diagonally similar to a positive child realization, with a nonzero real diagonal gauge normalized at its root. Attach the positive reference as above. On child component c, multiply its normalized gauge by M_{co}/f_c, which is nonzero because the edge

product is positive. The outward boundary entry then matches M_{co}; the matching positive edge product forces the inward entry to match as well. The child blocks and root diagonal already match. Thus

$$M = DBD^{-1}, \qquad D_{oo} = 1, \quad D_{vv} \neq 0, \tag{14}$$

where B is a positive conservative root-leak realization. The singleton is immediate. This proves the construction for all tree sizes and permits signed gauges.

For the actual competitor A with leak at ℓ, relocate the positive reference B from o to ℓ using (8); call the resulting positive matrix B_ℓ. It has the same full pair and is invertible because $P(0) = \prod_i r_i > 0$. Equation (14) and the positive relocation give $A = HB_\ell H^{-1}$ with every diagonal h_v nonzero. Conservation determines this gauge. Indeed, if the two leaks are λ_A, λ_B at the same ℓ, their column sums imply

$$h^T B_\ell = -\lambda_A h_\ell e_\ell^T, \qquad \mathbf{1}^T B_\ell = -\lambda_B e_\ell^T.$$

Invertibility and $\lambda_B > 0$ show $h^T = (\lambda_A h_\ell/\lambda_B)\mathbf{1}^T$. Its ℓ coordinate, with $h_\ell \neq 0$, gives $\lambda_A = \lambda_B$ and h constant. Therefore $A = B_\ell$. Every arbitrary-leak real competitor is positive, with no division by an unproved nonzero competitor rate or leak.

For positive models, relocation along a fixed path is invertible: the reverse single-edge slide recovers all three coordinates in (8). It gives a bijection between the root-leak fiber and each prescribed-site fiber. Principal resultants, edge products and positive rates preserve regularity. Finally, distinct sites yield distinct actual matrices: the column-sum vector $-\lambda e_\ell^T$, with $\lambda > 0$, recovers the site and leak, while off-diagonal entries recover all rates. Summing the N disjoint fixed-site fibers proves the unknown-site count.

5 Actual continuous-input observations

For the target (11), write

$$g_G(t) = \sum_{i=1}^{N} w_i e^{-r_i t}, \qquad y_G(t) = \int_0^t g_G(t-\tau)u(\tau)\,d\tau \quad (t \geq 0).$$

The target zero-initial output below means precisely y_G.

Proposition 5 (History interpretation)**.** *For the tree and target of Theorem 4, equality with the target zero-initial output for every continuous real input is equivalent to equality of the collocated transfer response. The same exhaustive real fiber and site counts therefore apply to these actual ODE histories. Positive leak relocation preserves every zero-initial collocated history at any fixed observer. This preservation itself does not need a full-mode target.*

Proof. For any real finite matrix A and continuous input, the actual solution is

$$x(t) = e^{tA}x(0) + \int_0^t e^{(t-\tau)A}e_o u(\tau)\,d\tau.$$

This exists globally and is the unique classical solution. The observed zero-initial kernel is $g_A(t) = e_o^T e^{tA} e_o$. Equal transfer functions imply equal kernel moments at zero by expanding the resolvent at infinity. Since the matrix exponential is entire, all moments determine the same kernel and hence every convolution output. Conversely equal outputs for all continuous inputs include the constant input one; differentiating its integral output gives equal kernels for $t \geq 0$. Their Laplace transforms agree for sufficiently large real s, giving equal rational responses. This reasoning requires no competitor stability or positivity and has actual solutions, so the quantified history predicate is not vacuous. Theorem 4 applies.

For relocation, normalize its positive diagonal similarity by $D_{oo} = 1$. Then $De_o = e_o$ and $e_o^T D = e_o^T$. It transports the forced equation, zero initial state and observed coordinate exactly, preserving every history. □

State observation at o is distinct from measuring discharge λx_ℓ at another leak vertex. The separately formalized collocated-discharge and signed-control results are not additional main claims here. In particular this paper does not assert arbitrary endpoint steering using nonnegative rainfall alone.

6 Proof correspondence, verification and limits

The accompanying `THEOREM_MAP.md` was prepared before this manuscript. It maps every main statement and necessary bridge to existing source declarations; no new theorem root is introduced by publication preparation. Table 1 gives compact navigation. Actual Lean types, including their hypotheses, are in the linked source and entrypoint index.

Paper scope	Existing module families
Theorem 1: all-real identity	LocatedAllRateJacobian; LocatedPolynomialWeight; LocatedExactMultiplicities; LocatedBoundarySingularSet
Exact powers and boundary	GraphResidualWitness; CoordinateMultiplicity
Corollary 2: original conjectures	CatenaryFormalPolynomial; FormalTreePolynomial; LocatedExactMultiplicities
Lemma 3 and root induction	FamilyGluing; FamilyResultants; RateChart; TreeJacobianRecurrence; AllTreeJacobian
Theorem 4: real fibers	RealFiberClassification; SupportedPositiveGauge; ConservationGaugeRigidity; RealLocatedClassification
Proposition 5: actual histories	RealLocatedODEClassification; MatrixTimeIdentification; LocatedLeakDynamics

Table 1: Navigation into the fixed existing Lean package, not a new audit.

The mathematical and public-scope source is fixed at `c2a2ce36ea6c5e000225574835f5a3ceddc748b4`. Lean is pinned to 4.34.1 (commit `5045d0056413266e57c625dcd7c365b10e377c52`); Mathlib is `d13f23b723b8a846827a245b89c10fc7d3f11612`. The original independent GitHub Actions run 38383557356 reports 220 owned modules, 1,677 exported roots and 53,370 used declarations, with trust-level-zero empty-kernel replay. Its allowed axioms are `Classical.choice`, `Quot.sound` and `propext`; 91 deliberately false source mutations were rejected. It starts with zero owned objects and builds the owned source afresh; this is not a claim that all of Mathlib was recompiled.

On 14 October 2026, publication preparation independently compared all 235 proof, reproduction and scope inputs against the current tree, the original source commit and the verified-source archive; all matched. The original ZIP digest was also matched to the GitHub API artifact metadata, and the corresponding job was confirmed successful. This is a source/archive/API readback of an existing independent run, not a new build or a newly expanded kernel audit. The preserved package provides its exact logs, lock manifests, negative controls and reproduction script. Our manuscript review is an internal assistant review, not external human peer review.

The full-target assumptions exclude repeated poles, fewer-mode data and complex-parameter fibers. The algebraic Jacobian theorem permits signed and zero rates but retains the finite single-leak tree and collocated input/state-observation model. General cyclic graphs, multiple leaks, off-diagonal input/output, noise, nonlinear catchments and predictive performance are outside this manuscript. The results do not determine unknown tree topology, nor claim a new topology-count corollary.

Data and proof availability. The fixed proof package and receipt are available in the author's repository, under `formalizations/hydrology-tree-core` and `verification/hydrology-tree-core-c2a2ce36`. This paper's source, theorem map, review and preparation checks accompany it under `preprints/finite-tree-jacobian-2026-10`. The paper restates and organizes the existing finite-tree result, rather than attributing that result to a new research round.

Declarations. The author received no external funding. OpenAI AI assistants assisted research, formalization, writing and internal review. Original material: copyright 2026 Yongxian Zhang, all rights reserved. Existing repository and third-party notices remain in force. This prepared manuscript does not itself assert an aiXiv submission, Release or DOI.

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